Chapitre 5 : Les tests d'hypothèses

Marie VAUGOYEAU (contact@marievaugoyeau.fr)

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1 Avant de commencer

Ce chapitre reprends les codes du chapitre 5 sur ***les tests d'hypothèses*** du livre **Langage R et Statistiques** : *Initiation à l'analyse de données* publié par les éditions ENI (2$^{\text{ème}}$ édition).

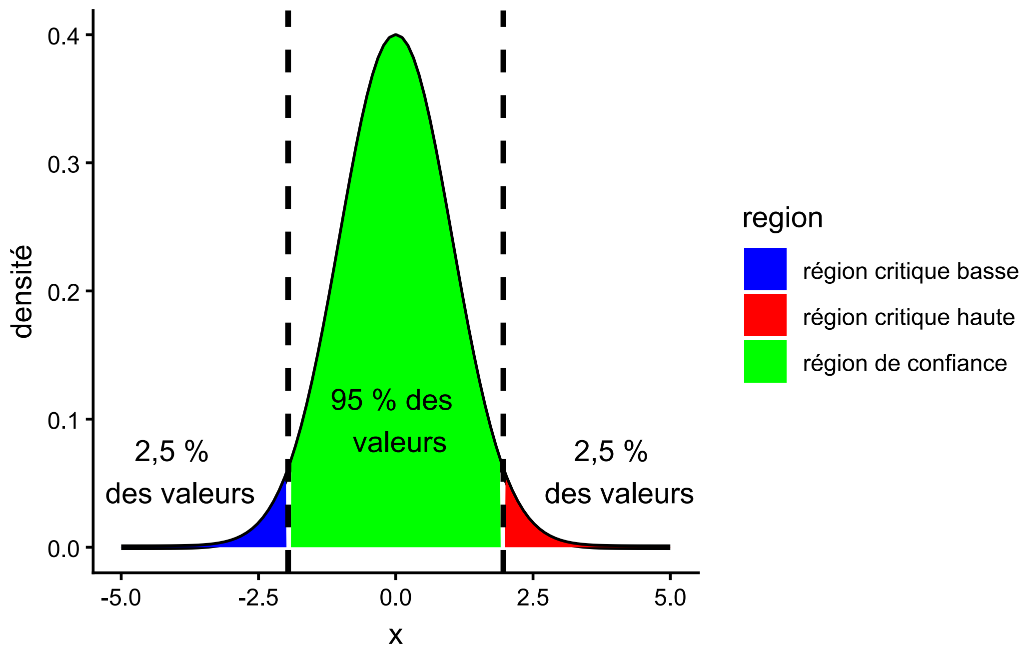
```
library(tidyverse)
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✔ dplyr     1.2.0     ✔ readr     2.1.6
✔ forcats   1.0.1     ✔ stringr   1.6.0
✔ ggplot2   4.0.2     ✔ tibble    3.3.1
✔ lubridate 1.9.5     ✔ tidyr     1.3.2
✔ purrr     1.2.1
── Conflicts ────────────────────────────────────────── tidyverse_conflicts()
──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors
```

2 Test bilatérale, unilatéral droit ou gauche

```
# choix de l'intervalle de représentation
intervalle <- seq(-5, 5, 0.1)

# test bilatéral
tibble(
  x = intervalle,
  "densité" = dnorm(intervalle, mean = 0, sd = 1),
  region =
    case_when(
      x < qnorm(0.025) ~ "région critique basse",
      x < qnorm(0.975) ~ "région de confiance",
      TRUE ~ "région critique haute"
    )
) |>
  ggplot() +
  aes(x = x, y = `densité`) +
  geom_line(linewidth = 1) +
  geom_area(aes(fill = region)) +
  scale_fill_manual(values = c("blue", "red", "green")) +
  geom_vline(xintercept = qnorm(0.025), linewidth = 1, linetype = "dashed") +
  geom_vline(xintercept = qnorm(0.975), linewidth = 1, linetype = "dashed") +
```

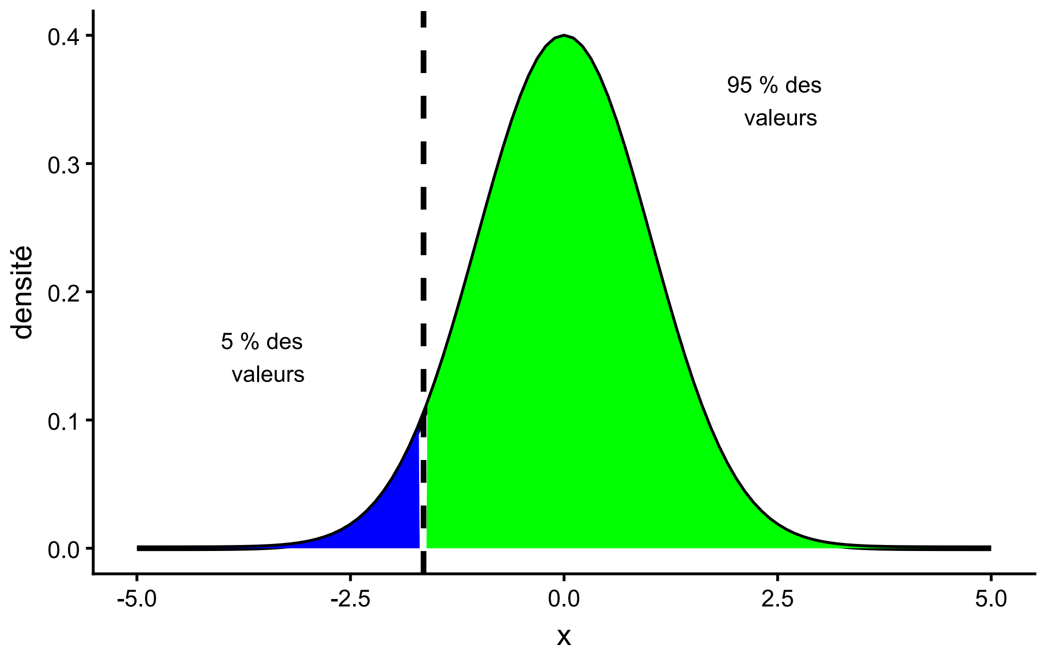
```
  annotate("text", x = -4, y = 0.06, label = "2,5 % \n des valeurs") +
  annotate("text", x = 4, y = 0.06, label = "2,5 % \n des valeurs") +
  annotate("text", x = 0, y = 0.1, label = "95 % des \n valeurs") +
  theme_classic()
```




```
ggsave("img/05Rl01N.png", width = 14, height = 6, units = "cm")

# test unilatéral à gauche
tibble(
  x = intervalle,
  "densité" = dnorm(intervalle, mean = 0, sd = 1),
  region =
    case_when(
      x < qnorm(0.05) ~ "région critique basse",
      TRUE ~ "région de confiance"
    )
)  |>
  ggplot() +
  aes(x = x, y = `densité`) +
  geom_line(size = 1) +
  geom_area(aes(fill = region)) +
  scale_fill_manual(values = c("blue", "green")) +
  geom_vline(xintercept = qnorm(0.05), linewidth = 1, linetype = "dashed") +
```

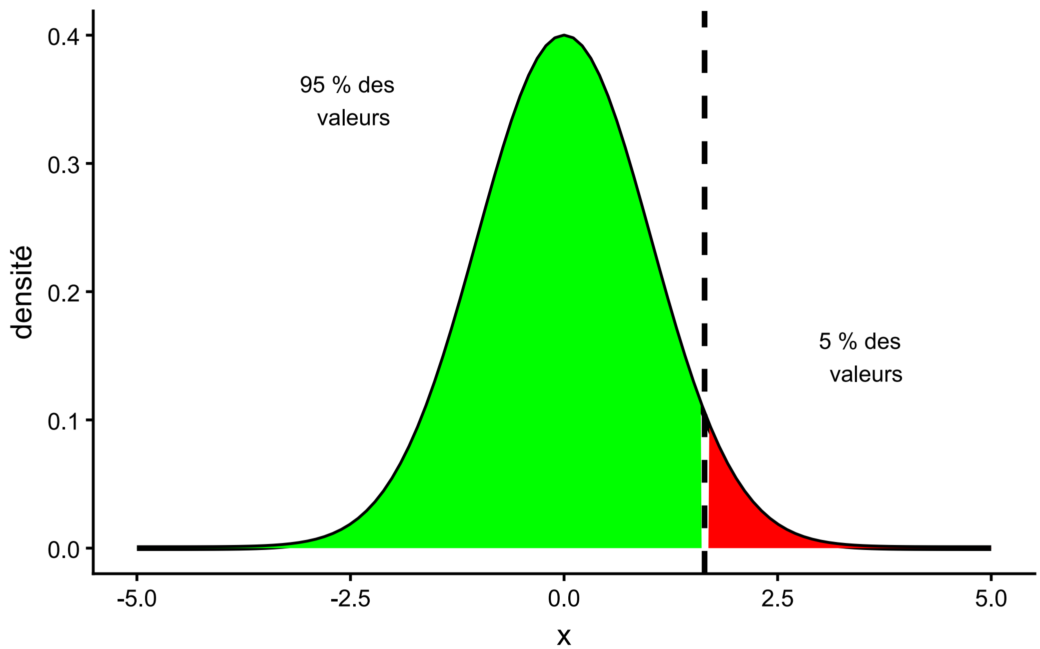
```
  annotate("text", x = -3.5, y = 0.15, label = "5 % des \n valeurs", size = 3) +
  annotate("text", x = 2.5, y = 0.35, label = "95 % des \n valeurs", size = 3) +
  theme_classic() +
  theme(legend.position = "none")
```

```
Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
ℹ Please use `linewidth` instead.
```




```
ggsave("img/05Rl01bisN.png", width = 6, height = 6, units = "cm")

# test unilatéral à droite
tibble(
  x = intervalle,
  "densité" = dnorm(intervalle, mean = 0, sd = 1),
  region =
    case_when(
      x > qnorm(0.95) ~ "région critique haute",
      TRUE ~ "région de confiance"
    )
)  |>
  ggplot() +

  aes(x = x, y = `densité`) +
```

```
  geom_line(size = 1) +
  geom_area(aes(fill = region)) +
  scale_fill_manual(values = c("red", "green")) +
  geom_vline(xintercept = qnorm(0.95), linewidth = 1, linetype = "dashed") +
  annotate("text", x = 3.5, y = 0.15, label = "5 % des \n valeurs", size = 3) +
  annotate("text", x = -2.5, y = 0.35, label = "95 % des \n valeurs", size =
3) +
  theme_classic() +
  theme(legend.position = "none")
```




```
ggsave("img/05Rl01terN.png", width = 6, height = 6, units = "cm")
```

3 Test de Shapiro-Wild

limite à 5000

```
shapiro.test(rnorm(10000, mean = 5, sd = 3))
```

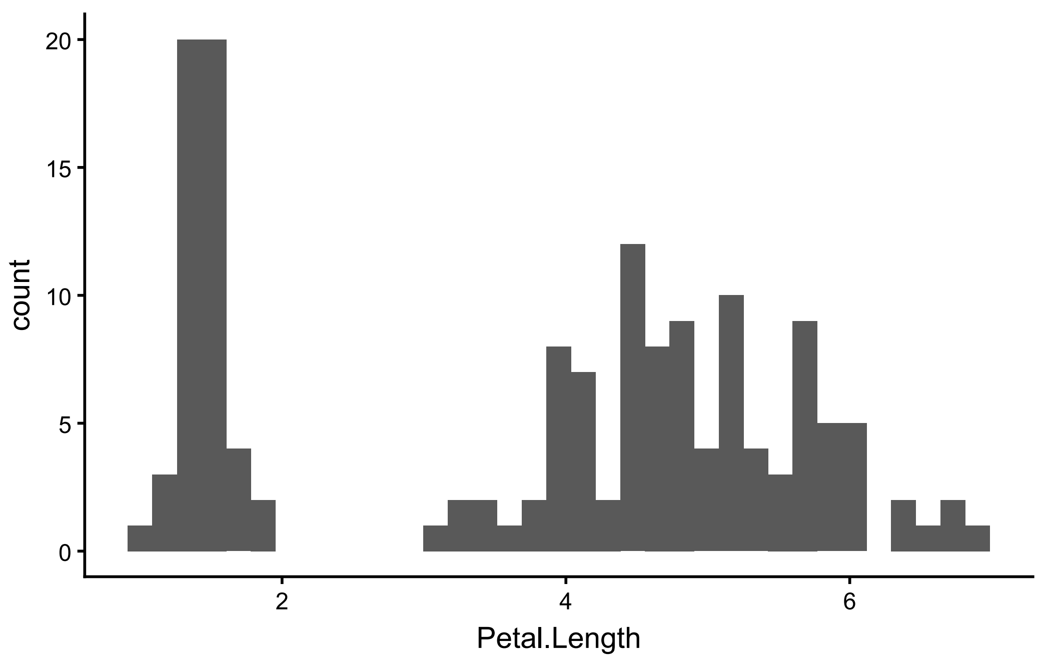
```
Error in `shapiro.test()`:
! la taille de l'échantillon doit être comprise entre 3 et 5000
```

exemple réalisation test de Shapiro

```r
shapiro.test(iris$Petal.Length)
```

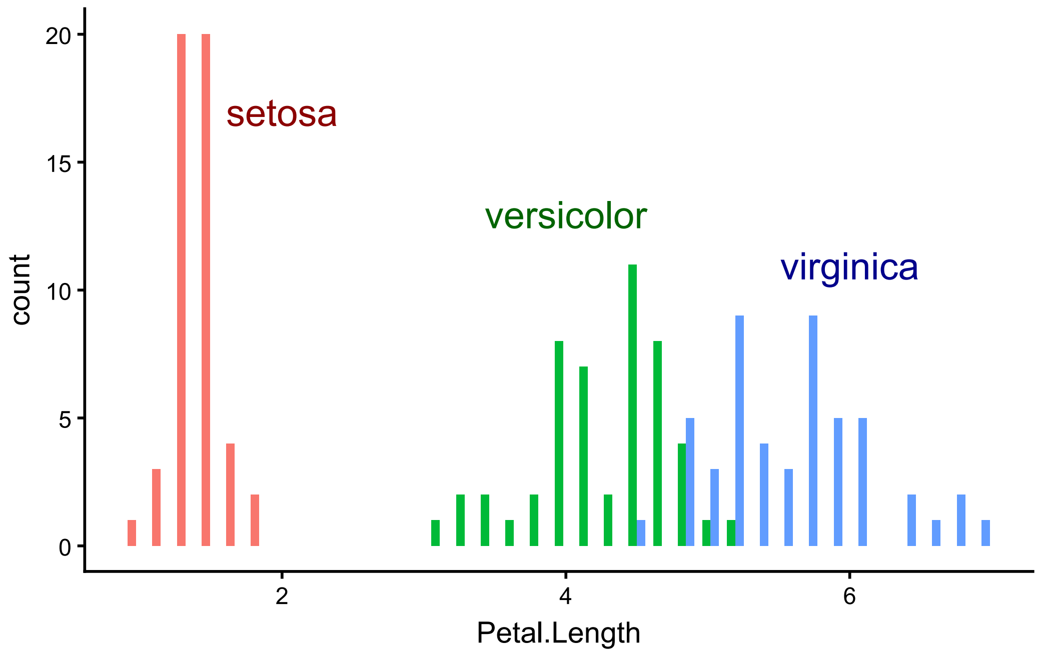
```
	Shapiro-Wilk normality test

data:  iris$Petal.Length
W = 0.87627, p-value = 7.412e-10
```

```r
ggplot(iris) +
  aes(x = Petal.Length) +
  geom_histogram(bins = 35) +
  theme_classic()
```




```r
ggplot(iris) +
  aes(x = Petal.Length, fill = Species) +
  geom_histogram(bins = 35, position = "dodge") +
  annotate("text", x = 2, y = 17, size = 5, label = "setosa", color = "darkred") +
   annotate("text", x = 4, y = 13, size = 5, label = "versicolor", color =
"darkgreen") +
   annotate("text", x = 6, y = 11, size = 5, label = "virginica", color =
"darkblue") +
```

```r
  theme_classic() +
  theme(legend.position = "none")
```




```r
ggsave("img/05Rl04N.png", width = 14, height = 10, units = "cm")
```

```r
shapiro.test((iris |> filter(Species == "setosa"))$Petal.Length)
```

```

    Shapiro-Wilk normality test

data:  (filter(iris, Species == "setosa"))$Petal.Length
W = 0.95498, p-value = 0.05481
```

```r
shapiro.test((iris |> filter(Species != "setosa"))$Petal.Length)
```

```

    Shapiro-Wilk normality test

data:  (filter(iris, Species != "setosa"))$Petal.Length
W = 0.99099, p-value = 0.7445
```

```
shapiro.test((iris |> filter(Species == "versicolor"))$Petal.Length)
```

```

	Shapiro-Wilk normality test

data:  (filter(iris, Species == "versicolor"))$Petal.Length
W = 0.966, p-value = 0.1585
```

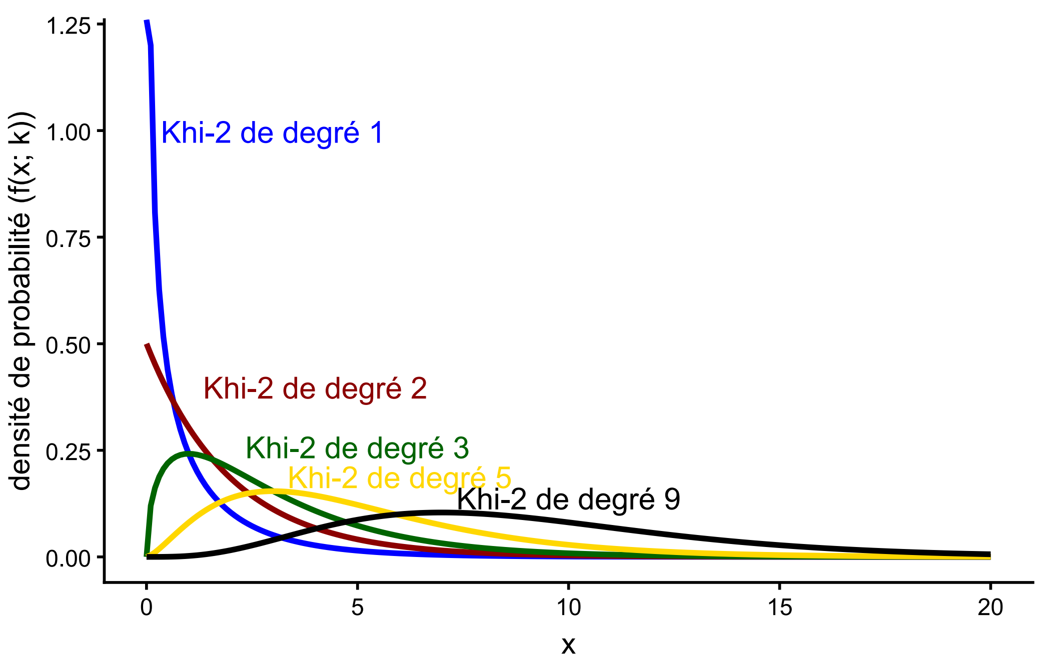
```
shapiro.test((iris |> filter(Species == "virginica"))$Petal.Length)
```

```

	Shapiro-Wilk normality test

data:  (filter(iris, Species == "virginica"))$Petal.Length
W = 0.96219, p-value = 0.1098
```

4 Loi Khi-2

```
# choix de l'intervalle de représentation
intervalle <- seq(0, 20, 0.1)

# loi khi-deux de paramètres k égale à 1, 2, 3, 5 ou 9
tibble(
  loi_khi_deux =
    rep(
      c(
        "k = 1",
        "k = 2",
        "k = 3",
        "k = 5",
        "k = 9"
      ),
      each = length(intervalle)
    ),
  x = rep(intervalle, 5),
  densite =
    c(
      dchisq(intervalle, df = 1),
      dchisq(intervalle, df = 2),
      dchisq(intervalle, df = 3),
      dchisq(intervalle, df = 5),
```

```r
      dchisq(intervalle, df = 9)
    )
) |>
  ggplot() +
  aes(x = x, y = densite, color = loi_khi_deux) +
  geom_line(size = 1) +
  theme_classic() +
 scale_color_manual(values = c("blue", "darkred", "darkgreen", "gold", "black"))
+
   annotate("text", x = 3, y = 1, size = 4, color = "blue", label = "Khi-2 de
degré 1") +
  annotate("text", x = 4, y = 0.4, size = 4, color = "darkred", label = "Khi-2
de degré 2") +
 annotate("text", x = 5, y = 0.26, size = 4, color = "darkgreen", label = "Khi-2
de degré 3") +
  annotate("text", x = 6, y = 0.19, size = 4, color = "gold", label = "Khi-2 de
degré 5") +
  annotate("text", x = 10, y = 0.14, size = 4, color = "black", label = "Khi-2
de degré 9") +
  ylab("densité de probabilité (f(x; k))") +
  theme(legend.position = "none")
```




```r
ggsave("img/05Rl06N.png", width = 14, height = 10, units = "cm")
```

4.1 Test d'adéquation du khi-2

```
chisq.test(
  x =  c(86, 111, 107, 91, 105, 100),
  p = rep(1/6, 6)
)
```

```

    Chi-squared test for given probabilities

data:  c(86, 111, 107, 91, 105, 100)
X-squared = 4.72, df = 5, p-value = 0.451
```

```
chisq.test(
  x =  c(86,    111, 107,   91, 105, 100)
)
```

```

    Chi-squared test for given probabilities

data:  c(86, 111, 107, 91, 105, 100)
X-squared = 4.72, df = 5, p-value = 0.451
```

```
chisq.test(
  x = c(4,2)
)
```

```
Warning in chisq.test(x = c(4, 2)): L'approximation du Chi-2 est peut-être
incorrecte
```

```

    Chi-squared test for given probabilities

data:  c(4, 2)
X-squared = 0.66667, df = 1, p-value = 0.4142
```

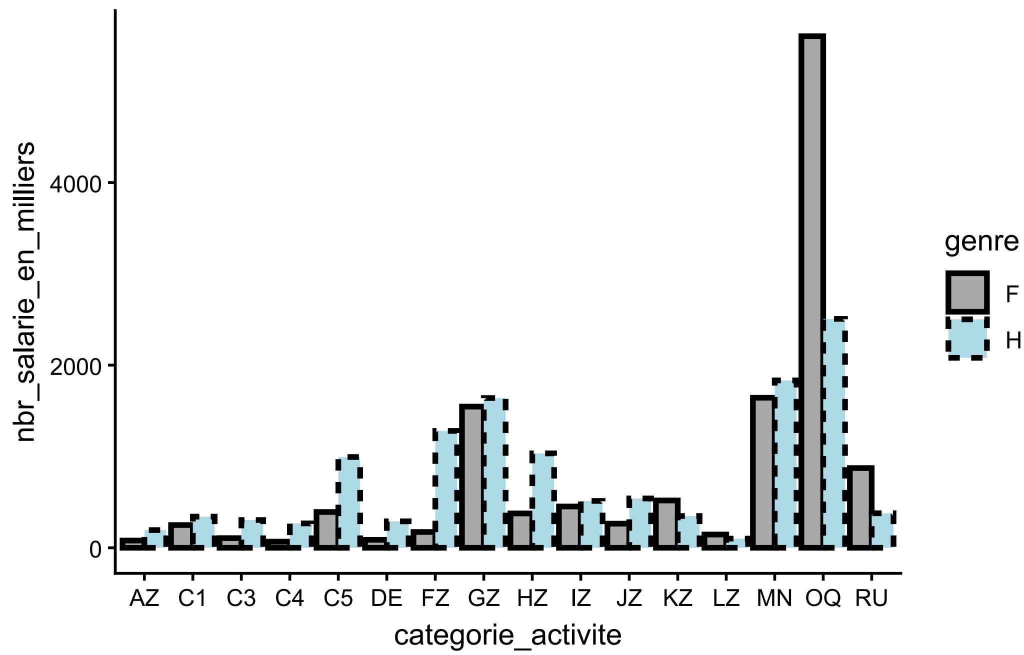
4.2 Test d'indépendance du Khi-2

création du tableau de données

```r
table_genre_naf <-
  tibble(
    categorie_activite = rep(c("AZ", "C1", "C3", "C4", "C5", "DE", "FZ", "GZ",
"HZ", "IZ", "JZ", "KZ", "LZ", "MN", "OQ", "RU"), 2),
    genre = rep(c("F", "H"), each = 16),
    nbr_salarie_en_milliers = c(79, 249, 106, 67, 392, 87, 173, 1546, 376, 452,
264, 518, 145, 1643, 5603, 873, 194, 341, 304, 267, 994, 290, 1280, 1640, 1034,
513, 540, 348, 100, 1833, 2507, 378)
  )
```

création du graphique

```r
ggplot(table_genre_naf) +
  aes(
    x = categorie_activite,
    y = nbr_salarie_en_milliers,
    fill = genre,
    color = genre,
    linetype = genre
    ) +
  geom_bar(
    stat = "identity",
    position = "dodge",
    linewidth = 1,
    color = "black"
    ) +
  scale_fill_manual(values = c("darkgrey", "lightblue")) +
  theme_classic()
```




```r
ggsave("img/05Rl10N.png", width = 14, height = 7, units = "cm")
```

```r
data.frame(
  Femmes = c(79, 249, 106, 67, 392, 87, 173, 1546, 376, 452, 264, 518, 145, 1643,
5603, 873),
  Hommes = c(194, 341, 304, 267, 994, 290, 1280, 1640, 1034, 513, 540, 348, 100,
1833, 2507, 378),
  row.names = c("AZ", "C1", "C3", "C4", "C5", "DE", "FZ", "GZ", "HZ", "IZ", "JZ",
"KZ", "LZ", "MN", "OQ", "RU")
) |>
  chisq.test()
```

```

	Pearson's Chi-squared test

data:  data.frame(Femmes = c(79, 249, 106, 67, 392, 87, 173, 1546, 376,     452,
264, 518, 145, 1643, 5603, 873), Hommes = c(194, 341,      304, 267, 994, 290,
1280, 1640, 1034, 513, 540, 348, 100,     1833, 2507, 378), row.names = c("AZ",
"C1", "C3", "C4", "C5",     "DE", "FZ", "GZ", "HZ", "IZ", "JZ", "KZ", "LZ", "MN",
"OQ",     "RU"))
X-squared = 3330.6, df = 15, p-value < 2.2e-16
```

4.3 Test d'homogénéité

```
library(rvest)
```

```


Attachement du package : 'rvest'
```

```
L'objet suivant est masqué depuis 'package:readr':

    guess_encoding
```

```
html <- read_html("https://www.insee.fr/fr/statistiques/4768237")

prop_cadre_genre <- html |>
  html_element("table") |>
  html_table(dec = ",") |>
  as_tibble(.name_repair = make.names)

prop_cadre_genre
```

```
# A tibble: 38 × 4
       X Femmes Hommes Ensemble
   <int>  <dbl>  <dbl>    <dbl>
 1  1982    4     10.3      7.8
 2  1983    4.4   11        8.3
 3  1984    4.5   11.3      8.5
 4  1985    4.6   11.2      8.5
 5  1986    5.2   11.7      9
 6  1987    5.5   12        9.2
 7  1988    5.9   12.7      9.8
 8  1989    5.9   12.4      9.7
 9  1990    6.7   13.2     10.4
10  1991    7.3   13.8     11
# ℹ 28 more rows
```

```
# exemple pour la sixième table
(read_html("https://www.insee.fr/fr/statistiques/4768237") |>
  html_elements("table"))[[3]] |>
  html_table(dec = ",")
```

```
# A tibble: 8 × 4
  ``                                                            Femmes Hommes Ensemble
  <chr>                                                          <dbl>  <dbl>    <dbl>
1 Professions libérales                                           47.4   52.6      100
2 Cadres de la fonction publique                                  50.4   49.6      100
3 Professeurs, professions scientifiques                          55.5   44.5      100
4 Professions de l'information, des arts et des spectacl…         44.6   55.4      100
5 Cadres administratifs et commerciaux d'entreprise               49.3   50.7      100
6 Ingénieurs et cadres techniques d'entreprise                    23.2   76.8      100
7 Ensemble des cadres et professions intellectuelles sup…         42.2   57.8      100
8 Ensemble des personnes en emploi                                48.5   51.5      100
```

```
# test du KLhi-2 d'homogénéité
prop_cadre_genre |>
  rename(annee = X) |>
  filter(annee > 1999) |>
  select(c(Femmes, Hommes)) |>
  chisq.test()
```

```

    Pearson's Chi-squared test

data:  select(filter(rename(prop_cadre_genre, annee = X), annee > 1999),
c(Femmes, Hommes))
X-squared = 0.72221, df = 19, p-value = 1
```